Executable Evidence Management for CAD-Guided Aircraft Inspection

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Abstract

Inspection evidence must remain aligned with the requirement, computation and output it supports. We develop an executable evidence interface for CAD-guided aircraft inspection that combines typed quality requests, producer dependencies, calibrated event bounds and authenticated output recovery. Event containment reuses stronger evidence, while exact-event fallback preserves the support of direct renewal. A provenance-bound repair transaction restores a corrected output without erasing its audit history. A controlled aircraft-skin study executes 3,072 outputs from 512 independently sampled scenes. All complete methods admit 41 of 48 service contexts; independent held-out simultaneous risk bounds for these services remain below 0.075. Containment reduces evidence-record evaluations by 38.10% relative to exact dependency caching. Measured stage caching reduces source-renewal CPU by 56.26%, shared by the matched dependency-based implementations. Validated repair restores 1,536 distinct eligible source/request contexts that an irreversible latch blocks. A separate 72-image path-traced study tests the unchanged front end on geometric fasteners and reflective surfaces. These experiments establish an explicit workflow for retaining, renewing and recovering engineering evidence through component and output changes.

Keywords: Executable operating contracts, Evidence reuse, Engineering change management, Risk calibration, Aircraft inspection

1. Introduction

An aircraft inspection image becomes useful when its observations can be located on the airframe and assessed against an engineering requirement. CAD-guided aircraft inspection uses model features to establish this spatial correspondence [1]; fastener-inspection systems then evaluate surface components under the available imaging conditions [2]. The resulting record must retain the conditions under which its quality was established. A change in exposure can alter feature extraction while leaving the CAD geometry intact. A relaxed mapping tolerance can make an existing result acceptable. Correcting an attachment error can restore the association between an observation and its verified output.

These changes require selective maintenance of both computations and their supporting evidence. Requirement-based digital-twin studies identify context, uncertainty, thresholds and historical information as necessary inputs to maintenance decisions [3]. Selective sensor calibration similarly concentrates effort on measurements identified as unreliable [4]. For inspection, the corresponding decision is which numerical outputs and quality statements remain applicable after a requirement, producer component or output association changes. Recomputing the complete pipeline spends work on unaffected stages; retaining a statistical statement without checking its event and producer can attach it to an incompatible result.

Existing research supplies substantial parts of this decision. CAD consistency analysis identifies valid repairs after model edits [5], lifecycle knowledge graphs preserve specification, version and operating-state histories [6], and dynamic safety cases update evidence and arguments when monitored conditions change [7]. Provenance models describe how an output was generated and revised [8]. The question studied here is how these records support an executable inspection decision: whether a calibrated event still covers the requested quality, whether the functions defining that event remain compatible, and whether the evidence is attached to the authenticated current output.

We develop an interface that makes these three conditions jointly checkable. The first contribution is a request-to-event compilation and renewal rule: event containment reuses a stronger statement, while exact-event fallback preserves every service admitted by direct renewal under aligned calibration. The second is a recovery protocol that couples statistical admissibility to a versioned output binding; a verified repair restores the binding and preserves the rejected attempt in its history. The third is a controlled evaluation that separates evidence preparation from numerical caching and checks the same decisions through independent rules and registry implementations. A paired image study, held-out risk checks and a separate geometric-rendering study establish the behavior under requirement, producer and output changes. Figure 1 shows the engineering context; the running example in Fig. 2 follows one observation through these decisions.

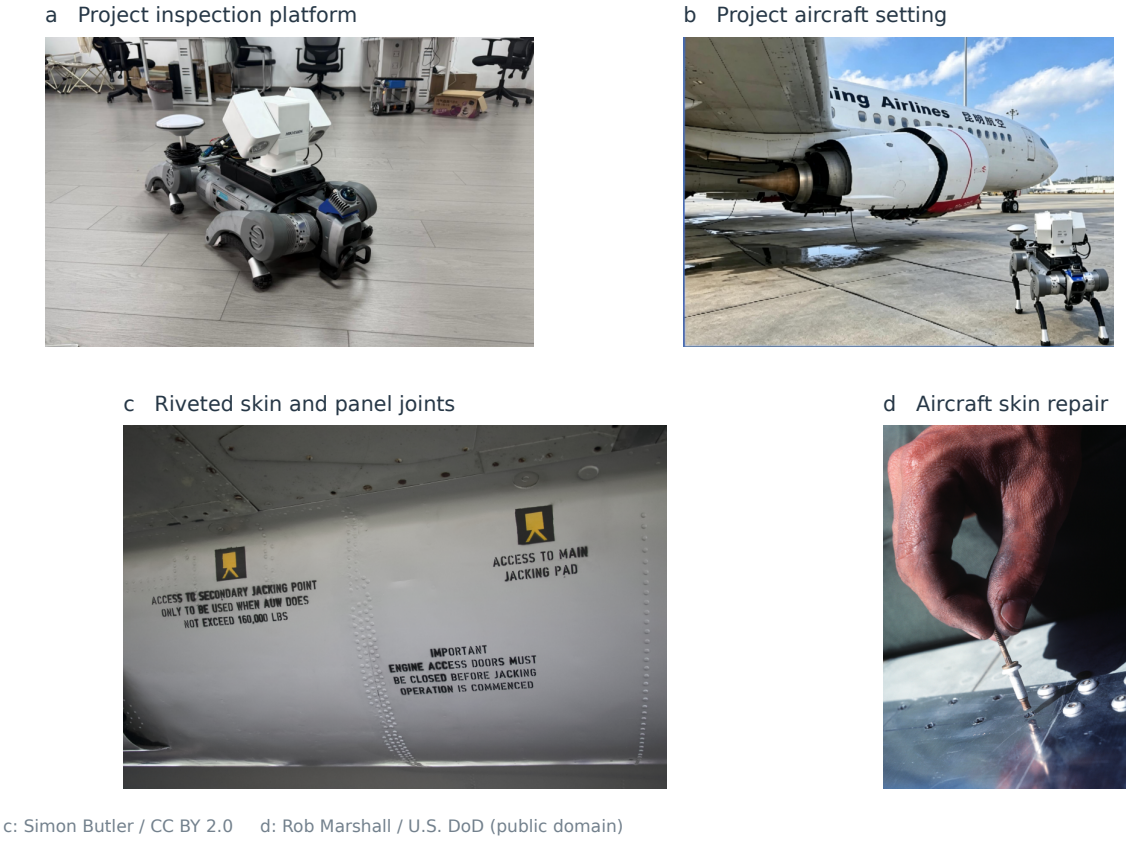


Figure 1: Aircraft inspection at platform and surface scales. (a,b) Original project photographs. (c) Under-fuselage panels on Avro Vulcan XM655, photographed by Simon Butler (CC BY 2.0). (d) Aircraft sheet-metal repair, photographed by Rob Marshall (U.S. Department of Defense, public domain). Sources and image credits are archived in the Supplementary Material.

2. Related work and comparison scope

2.1. Engineering knowledge and change-dependent decisions

Engineering knowledge representations connect requirements, model entities and decisions. Petnga and Austin integrate ontological reasoning with physical quantities and unit-aware computations [9]. Zou and Feng resolve inconsistencies introduced by direct CAD editing through constraint analysis and ranked repair candidates [5]. These approaches establish that an engineering representation should support an actionable consistency decision. Quinn and McArthur evaluate representational completeness and expressiveness through

industrial concept mappings and application relations [10]; inspection-robot ontologies similarly relate task requirements to hardware capabilities [11].

Recent AEI systems extend the connection to executable models and lifecycle histories. Xiao et al. map design and building-model information into executable energy-analysis models [12]. Yang et al. separate persistent entities, specifications, released versions and runtime states, with explicit change actions and retained histories [6]. We use the same engineering distinction between identity and changing state. The decision added here concerns a quality event: its units and thresholds determine which calibrated statement and producer functions are sufficient for the current request.

2.2. Provenance, selective recomputation and evolving evidence

The W3C PROV model describes generation, use, derivation, revision and invalidation; its constraints formalize consistency among these records [8, 13]. For computational workflows, PDIFF compares execution provenance to locate differences caused by data, service or workflow revisions [14]. RECOMP studies selective recomputation of earlier results as data and methods evolve [15], and version-aware restart trees identify affected outputs for recomputation [16]. Build-system theory supplies the dependency-directed retention of unchanged computations [17]. These results motivate a dependency cache as the numerical baseline and a paired regression gate for declared producer invariances.

Evidence maintenance also has an established assurance literature. Dynamic safety cases couple monitored variables, confidence structures and update rules, including removal of arguments whose assumptions become invalid [7]. SafeMachine organizes machine-safety assessment around traceable, context-aware evidence [18]. Assumption–guarantee contracts specify component obligations [19], and runtime-assurance architectures mediate permissible execution [20]. Our interface specializes evidence maintenance to a calibrated scene-quality event and the output receiving that statement. Its recovery transaction checks the trusted current binding; its event rule independently determines whether the requested service can resume.

2.3. Risk calibration and visual inspection

Learn then Test selects predictive programs using hypothesis tests with high-probability risk control [21]. Conformal Risk Control calibrates a monotone loss to control its expectation [22]. The present study uses exact binomial inversion [23] with a fixed finite-family allocation. Event inclusion then supplies the ordering needed for reuse; exact fallback reserves the original request's test and preserves its admissible support. This statistical mechanism gives a precise meaning to the confidence attached to an inspection service.

Industrial visual inspection also requires the evaluated event to reflect the task. Component-aware anomaly detection separates component structure and logical arrangement from local appearance [24]. CAD-guided aircraft inspection and rivet inspection provide the spatial and component-level setting for our capture-to-CAD requirements [1, 2]. Training on appearance-randomized renderings can improve transfer across image domains [25]; Xu et al. evaluate simulation-to-experiment adaptation for ultrasonic concrete-damage signals [26]. We therefore evaluate geometric-rendering transfer as a distinct image population. Numerical quality, statistical applicability and output provenance are measured as separate parts of the same inspection workflow.

Table 1: Closest research connections and the decision examined in this study. The established capabilities are complementary foundations.

Research connection	Established capability	Inspection decision examined here
Lifecycle knowledge [6]	Preserve identities, versions, states and change dependencies	Which requested quantities remain covered after a producer change?
Dynamic evidence [7]	Update arguments and evidence as monitored conditions change	Which scene-quality event has an applicable numerical risk bound?
Workflow provenance [14, 16]	Locate changed computations and affected outputs	Which statistical statements can accompany retained or corrected outputs?
Risk-controlled selection [21]	Admit programs under a calibrated risk allowance	How can event reuse preserve exact-renewal support?

Table 2: Matched comparisons and the question each resolves.

Comparison	Shared information	Evaluated distinction
Whole renewal / DAG cache	Producer and exact events	Required numerical recomputation
DAG / contract + fallback	Same numerical cache and risk family	Event containment and evidence work
Contract / independent rules	Same event and dependency semantics	Implementation agreement
Permanent latch / validated repair	Trusted output records	Recovery after a corrected binding
Repair / authoritative registry	Same provenance and transactions	Recovery semantics across implementations

3. Inspection requirements and executable evidence

3.1. A change-driven inspection example

Consider a camera observation used to associate fastener positions with CAD addresses. The complete-service request constrains spatial sampling, motion blur, correct-address coverage, fitted residual and physical mapping error. The baseline producer satisfies the calibrated request. Increasing exposure preserves the geometric sampling scale but changes motion blur and image formation. The new producer therefore renews the affected statistical statement. A relaxed request can still be supported even when the strict event fails its risk allowance.

The numerical output also carries a source identifier, producer signature and version. If an old output is attached after the update, the binding is rejected. Correcting that binding is a separate operation from changing the quality requirement: the registry checks the active producer, output digest and repair revision before service resumes. This example keeps three decisions distinct throughout the paper: numerical reuse, statistical admissibility and output attachment. For the first held-out source, geometric sampling remains 0.858 mm/pixel after a 50% exposure increase, while blur changes from 1.050 to 1.575 pixels. This particular image satisfies the complete quality predicate in both configurations. Across the calibration population, however, the strict complete-service upper bound changes from 0.097 to 0.272, exceeding the allowance of 0.15. The relaxed exact event has upper bound 0.080 and remains admissible. Thus a good individual output and support for a producer over its operating population answer different engineering questions.

3.2. Typed requests and producer dependencies

An engineering request is $R = (g, a, \varphi, \delta)$: population g, service a, quality predicate φ and allowable scene-level failure probability δ. Its grammar is

$$\varphi ::= \mathrm{Interval}(m, u, \ell, h) \mid \varphi \wedge \varphi, \tag{1}$$

where m identifies a quantity and u its engineering unit. Following unit-aware engineering knowledge models [9], compilation checks dimensions, converts units, intersects repeated bounds and canonicalizes the event. Nonfinite required values fail the quality predicate; an absent required field prevents evaluation as a schema error. The accepted set and its failure event are

$$S_R = \{z : \ell_{R,j} \le z_j \le h_{R,j},\ j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}. \tag{2}$$

The knowledge structure $\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma)$ stores typed entities, typed relations, immutable specifications and executable predicates. Its relation signatures include requires : Request $\to$ Event, bounds : Card $\to$ Event, dependsOn : Quantity $\to$ Provider and uses : Receipt $\to$ SourceVersion. Type checks reject incompatible endpoints. The compiler maps these records to the decisions in Table 3.

For quantity j, the registered provider manifest declares dependencies D_j; the request uses $D_R = \bigcup_{j \in J_R} D_j$. The producer projection contains the corresponding function versions and parameters. Each numerical stage also uses an ordinary content-addressed key, following dependency-directed computation [17],

$$h_v = \mathrm{Hash}\big(v, \sigma_v, \{h_u : u \in \mathrm{pa}(v)\}\big), \tag{3}$$

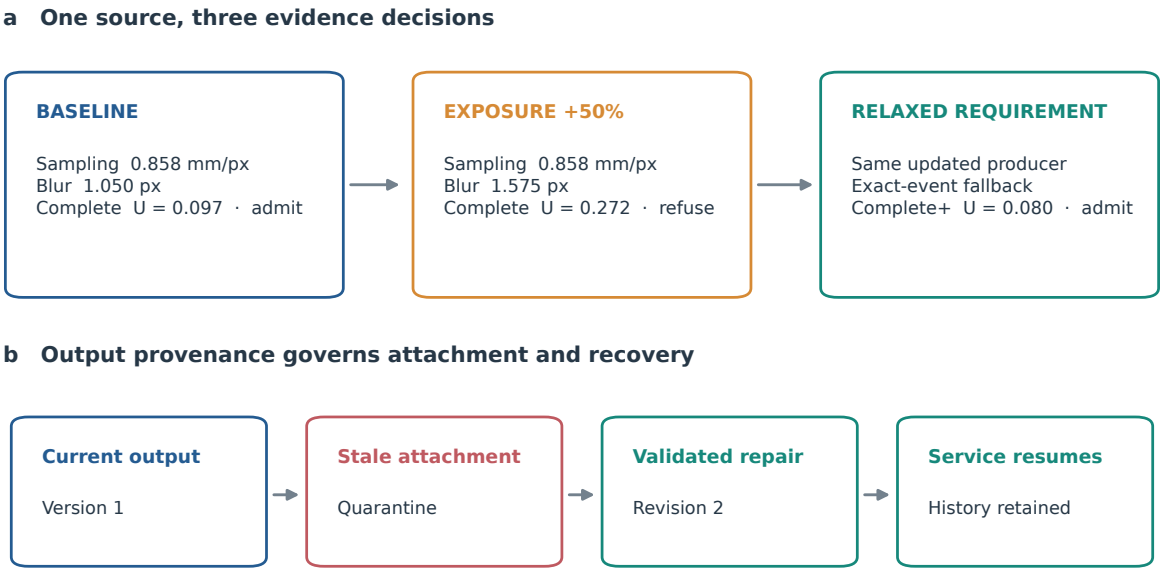


Figure 2: One inspection output through change and recovery. Exposure changes the requested blur and image quantities while preserving geometric sampling. The strict and relaxed statements use their own calibrated events. A stale attachment enters quarantine; a verified repair restores the current binding and keeps the rejected attempt in the audit record. Numerical values come from the first held-out source and the archived event cards.

which retains a cached value when the registered stage and its inputs agree. The request projection selects the functions relevant to its quality statement. A component-update regression gate re-executes common inputs and vetoes a declared invariance when an unaffected quantity changes. Runtime checks use the resulting registered signatures.

Table 3: Records connecting a quality requirement to an output.

Record	Required content	Decision
Request	Population, service, units, intervals, allowance	Compile event and dependencies
Evidence card	Event, producer, calibration lineage, n, k, U	Cover the requested event
Source version	Observation identity, active producer, output digest	Authenticate the current binding
Repair transaction	Expected revision, trusted output and new revision	Restore a quarantined binding
Execution record	Plan, attachment and transaction history	Explain issuance or refusal

3.3. Image-to-CAD quality

Pixel extraction finds candidate fastener centers, and a pose prior supplies coarse CAD projections under the perspective camera model [27]. Minimum-cost assignment uses image-plane distances and a prior gate, implemented with the modified Jonker–Volgenant assignment routine described by Crouse [28]. For assigned

correspondences $\mathcal{I}$, pose refinement solves

$$\widehat{\xi} = \arg\min_{\xi} \sum_{i\in\mathcal{I}} \sum_{d\in\{x,y\}} \rho\left(\frac{\pi_\xi(q_i)_d - u_{i,d}}{\sigma_u}\right) + \sum_{d=1}^{3} \left[\rho\left(\frac{\Delta t_d}{\sigma_t}\right) + \rho\left(\frac{\Delta\theta_d}{\sigma_\theta}\right)\right]. \tag{4}$$

Here q_i is a nominal CAD address, u_i a pixel-derived center, and π_ξ the perspective projection. Translation and local rotation perturb the prior. The componentwise Huber loss [29] has transition 2. Matching and fitting use the pixels, nominal CAD and pose prior.

Offline scoring distinguishes fitted residual from physical mapping error. Residual includes every assigned correspondence, including incorrect assignments. Mapping error evaluates the estimated image-to-CAD map $H_{\widehat{\xi}}$ at every planned physical image location:

$$e_{\mathrm{CAD}} = \max_{i\in\mathcal{P}} \|H_{\widehat{\xi}}(u_i^\star) - q_i^\star\|. \tag{5}$$

Scene geometry supplies these evaluation labels after estimation. The attachment gate authenticates output identity and provenance; physical quality labels enter calibration and evaluation.

4. Statistical reuse and validated recovery

4.1. Validity of selected services

For k failures among n independent complete calibration scenes, a one-sided Clopper–Pearson bound [23] is

$$U(k,n;\alpha) = \begin{cases} 1, & k = n, \\ \mathrm{Beta}^{-1}(1-\alpha; k+1, n-k), & k < n. \end{cases} \tag{6}$$

The controlled study fixes six producers and eight events before sampling, with $K = 48$, $\beta = 0.05$ and $\alpha = \beta/K$. The common confidence event covers all 48 risks, using the familywise selection principle of Learn then Test [21]. A plan selected from these bounds is committed before evaluating a new scene.

A stronger requirement provides evidence for a weaker one: $S_A \subseteq S_R$ implies $E_{p,R} \subseteq E_{p,A}$. Producer transfer additionally requires identity of the requested functions,

$$z_{p',j}(x) = z_{p,j}(x) \quad \text{for all } j \in J_R \text{ and supported } x. \tag{7}$$

Registered dependency projections implement this condition through the declared provider interface. The numerical regression gate tests that interface against paired executions.

Proposition 1 (Validity of calibration-selected services)**.** *Let $C \sim P^n$ be calibration data and let a new scene $X \sim P$ be independent of C. Fix K producer/event functions with one-sided allocations summing to at most β. With probability at least $1-\beta$ over C, every admitted request selected using C has fresh-scene risk at most δ, provided that its event is pointwise contained in an anchor event from this fixed family with $U_A(C) \le \delta$ and its transferred functions satisfy Eq. (7).*

Proof. The union bound gives a calibration event of probability at least $1-\beta$ on which every fixed-family risk is bounded by its $U_A(C)$. On that event, pointwise inclusion gives $\Pr_X(E_R \mid C) \le \Pr_X(E_A \mid C) \le U_A(C) \le \delta$ for every eligible selection. Requested-function identity preserves the inclusion after transfer. □

This proposition states the quality guarantee for a planned service. Runtime attachment supplies the provenance of the result using that service. For any subsequent issuance indicator I, the joint event satisfies $\Pr(E_R \cap \{I = 1\} \mid C) \le \delta$ on the same calibration event. The Supplementary Material gives the distinction between this bound and a risk conditioned on a scene-dependent selection gate.

4.2. Containment with exact-event fallback

For a committed producer, the interface first searches compatible covering cards with $U_A \leq \delta$. If none qualifies, it constructs the request's exact-event card from the fixed family. The route is

$$Q(R) = \begin{cases} \text{reuse}(A), & \mathcal{A}(R) \neq \varnothing, \\ \text{exact}(R), & \mathcal{A}(R) = \varnothing,\ U_R \leq \delta, \\ \text{refuse}, & \text{otherwise.} \end{cases} \tag{8}$$

Here $\mathcal{A}(R)$ contains cards with matching population, service, producer projection and containing events whose bounds satisfy $U_A \leq \delta$. Failed anchors and refusal cards count toward construction work.

Proposition 2 (Support preservation with exact fallback)**.** *Suppose the compared cards use the same n calibration scenes and allocation α, and transferred request quantities agree on those scenes. Containment followed by exact-event fallback admits exactly the requests admitted by the complete exact-event catalogue.*

Proof. Containment gives $k_R \leq k_A$. At fixed n and α, the binomial upper bound is nondecreasing in k, so admission by an anchor implies $U_R \leq U_A \leq \delta$. Otherwise, fallback evaluates the same exact event as the complete catalogue. □

The fixed family reserves the allocation for every exact fallback. A failed strict anchor can therefore lead to an admissible relaxed event without changing the confidence budget.

4.3. Quarantine and authenticated repair

Provenance distinguishes the generation, revision and invalidation of an output [8, 13]. Our output binding identifies a source, producer, version and payload digest. An authoritative registry stores the active tuple. Issuance requires both an admissible plan and agreement with that tuple. An invalid attachment places its binding workflow in quarantine and records the rejected attempt. Ordinary retries preserve that state.

Recovery uses an explicit transaction. It presents the expected registry revision, identifies the trusted active output, and requests a strictly newer revision. The registry verifies the producer, payload digest, source lineage and current calibration identity before accepting the transaction. A successful repair replaces the binding while retaining the audit history. A missing repair certificate, a wrong digest, a replayed transaction or an unregistered alias cannot create a valid repair. Invalid repair commands are rejected without overwriting a previously valid committed state.

The resulting invariant is direct: every issued binding equals a currently authenticated registry record, and every transition out of quarantine has a successful repair transaction in its history. Numerical quality and binding integrity have separate roles. A valid repair restores the association to a trusted output; the selected event card determines whether that producer supports the request. The comparison includes an authoritative-snapshot implementation and an independent event-log implementation, together with the earlier permanent latch. These controls quantify recoverability and check the same trust conditions through different implementations.

5. Experimental design

5.1. Controlled scene population and service requirements

The controlled image study draws 256 calibration and 256 held-out scenes from four layouts: parallel rows, curved rows, staggered fasteners and an access-panel perimeter. Each complete scene independently draws surface curvature, camera pose, material appearance and site perturbations. Layouts contain 28–36 sites, with fastener radii of 2.0–3.4 mm and typical spacing of 28–40 mm. Missing sites occur with probability 0.025; an edge occluder covers 0–10% of the raster. Shared and local physical-to-CAD displacement standard deviations are 0.4 and 0.15 mm.

Baseline resolution is 768×576 pixels. Focal length is 1,200–1,600 pixels, distance 0.70–1.15 m, exposure 3–7 ms and motion 0.06–0.18 m/s. Five changes independently increase focal length by 15%, exposure by 50%, expand the raster to 896×672, multiply illumination by 0.45, or reduce the decoder contrast threshold from 18 to 10. All six configurations execute on every scene, producing 3,072 numerical outputs. Local contrast and 5–240-pixel connected components supply candidate centers. A 30-pixel prior gate constrains minimum-cost CAD assignment; at least six matches trigger fitting with scales of 0.5 pixel, 3 mm and 0.001 rad.

Table 4 specifies the quality events used in both calibration and evaluation. These inherited benchmark requirements were fixed before sampling the new population. They describe capture-to-CAD quality at the modeled fastener scale. The primary allowance $\delta = 0.15$ applies to violation of the requested conjunction by a complete scene. A sweep from 0.02 to 0.20 reports how admission changes with this operating choice.

Table 4: Predefined capture-to-CAD requests. GSD is in mm/pixel, blur and residual in pixels, coverage in percent and mapping in mm. A dash denotes an unconstrained quantity.

Request	GSD $\leq$	Blur $\leq$	Coverage $\geq$	Residual $\leq$	Mapping $\leq$
Optical	1.2	3	—	—	—
Optical+	1.5	4	—	—	—
Acquisition	1	2	80	—	—
Acquisition+	1.2	3	60	—	—
Association	—	—	80	8	8
Association+	—	—	60	12	12
Complete	1	2	80	8	8
Complete+	1.2	3	60	12	12

A separate 128-scene stress population increases missingness to 0.14, occluded area to 0–35%, shared CAD displacement to 2 mm and local displacement to 0.6 mm. Its distribution identity remains separate from the calibrated population. The first held-out controlled input and independently rendered surfaces appear in Fig. 3.

5.2. Geometric rendering and front-end transfer

Path tracing evaluates light transport through the interaction of geometry, materials and illumination [30]. A separate Blender 4.3.2 Cycles study generates 24 independently seeded panels, each with baseline, exposure-gain and reduced-illumination configurations. Curved panel meshes, overlapping joints, shallow three-dimensional fastener heads and countersink rings form the surface. Painted and metallic materials use sampled roughness and surface variation; three area lights provide diffuse and reflected illumination. Four development seeds establish rendering settings before the held-out sources are generated.

The renderer produces 768×576 images with at most 32 samples per pixel, adaptive sampling threshold 0.01 and no denoising. The gain update adds 1.3 stops; the illumination update scales light and world radiance by 0.1. The camera is static, so this gain intervention changes brightness without a shutter-motion model. All 72 images pass through the unchanged contrast detector and pose estimator. An independent Blender camera projection is checked against the numerical projector. Physical sites enter evaluation after matching and fitting. This renderer defines a separate transfer population; its quality outcomes are evaluated directly.

5.3. Matched renewal and independent risk checks

Whole renewal recomputes the full producer and builds exact-event cards. The DAG baseline reuses unchanged stages and identical events. The contract uses the same DAG with containment and exact fallback. Independent rules implement the same interval, producer and bound predicates. All methods share the numerical estimator, calibration inputs and confidence allocation. Catalogue construction includes every evaluated event, including unsuccessful anchors.

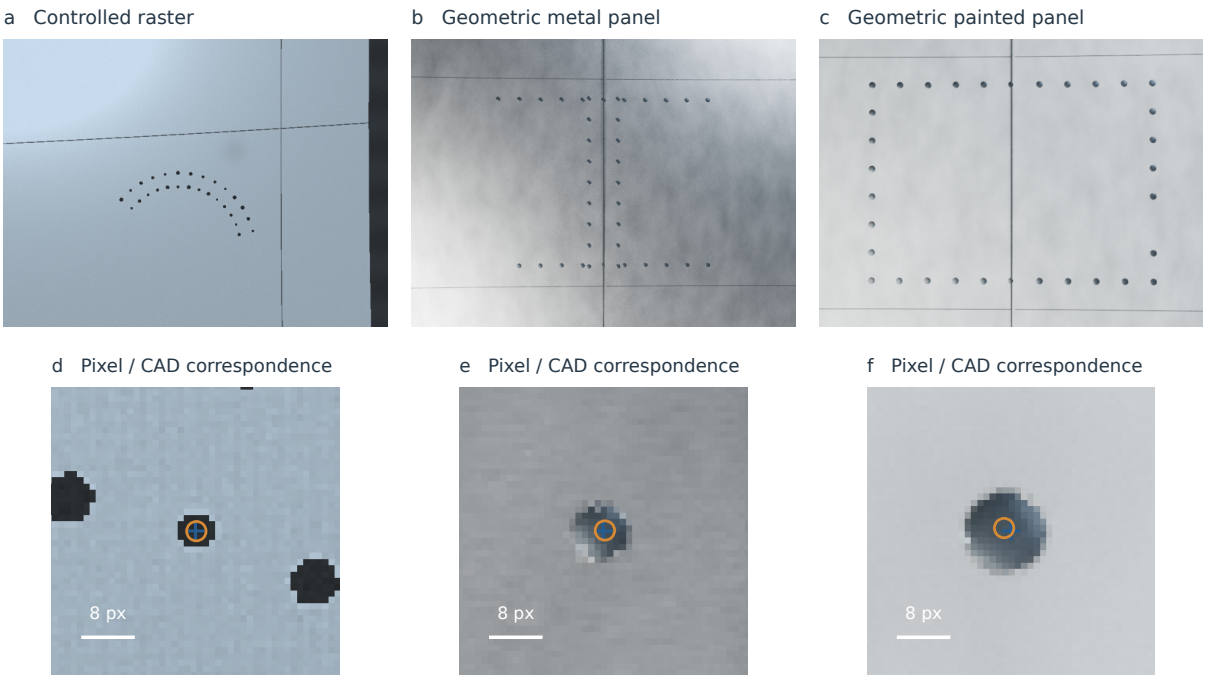


Figure 3: Images actually processed by the two producers. Controlled raster inputs support the large paired calibration study. Independently generated path-traced panels use geometric fastener heads, reflective materials and area illumination. The overlays show projected CAD locations and centers extracted from image pixels. Panels follow fixed source-order selection; their source identities and full-resolution images are archived.

Source-renewal timing uses 16 independent inputs, five updates and eight requests, giving 640 paired cases per method. Each case supplies the same baseline cache, rotates method order and checks numerical agreement. Process CPU and elapsed time include required stages and the scalar event check. Catalogue preparation is measured separately over the 256 stored calibration records in 11 repetitions. Execution uses Python 3.12 and SciPy [31] on a Linux AMD EPYC 9V74 host with one numerical thread per worker. Pose refinement uses its trust-region reflective least-squares solver with the specified Huber loss.

Plans are fixed using the calibration split. Independent held-out risk checks then apply the same 48-event allocation to the test failure counts; these bounds assess the planned services. The offline component regression study uses 64 additional inputs and three conditional hidden couplings: ROI/focal length, illumination/exposure and decoder/focal length. Clean updates execute on the same inputs.

5.4. Recovery and independently checked output bindings

The recovery study replays the 256 held-out image sources under eight requests and two binding faults, yielding 2,048 distinct source/request contexts and 4,096 fault scenarios. One fault attaches a stored baseline output after the exposure update; the other attaches another source's stored output. All substituted records differ numerically from the intended output. The authenticated payload contains the archive locator, fitted residual and matched/planned counts. Physical mapping and coverage labels are reserved for independent scoring.

Each scenario follows 13 steps: baseline and updated registration, wrong attachment, retry, unregistered alias, five malformed repair attempts, valid repair, duplicate repair and resumed request. Repair evidence checks the expected revision, active producer, source, payload digest and calibration identity. The comparator derives numerical origin from the immutable archive records. Permanent latch, authoritative snapshot,

validated repair, independent event-log replay and a metadata-only ablation receive the same records. Recovery availability counts only requests admitted for the updated producer.

6. Results

6.1. Calibrated services and independent held-out risk

The controlled held-out images have mean correct-address coverage 96.14%. Refinement reduces median prior translation error from 4.328 to 1.149 mm. Median fitted residual is 0.639 pixel, and median and 95th-percentile mapping errors are 0.502 and 1.052 mm.

At allowance 0.15, calibration admits 41 of 48 producer/request contexts (Fig. 4). Exposure removes strict acquisition and complete-service support; lower light additionally affects association. All complete methods give the same decisions. Anchor-only routing admits 38 contexts. Exact fallback restores relaxed acquisition and complete service after exposure, and relaxed acquisition under lower light.

The separate held-out check evaluates all 48 fixed programs on all 256 test scenes. Among the 41 calibration-admitted services, the largest test failure count is 7/256. Their simultaneous one-sided test bounds are all at most 0.07457, using a separate 95% family statement with allocation 0.05/48. These checks use the complete test population and leave the selected plans unchanged.

The allowance sweep admits 0, 18, 40, 41 and 43 contexts at 0.02, 0.05, 0.10, 0.15 and 0.20, respectively. Every complete method preserves this support. The Supplementary Material reports the corresponding construction work and all held-out bounds.

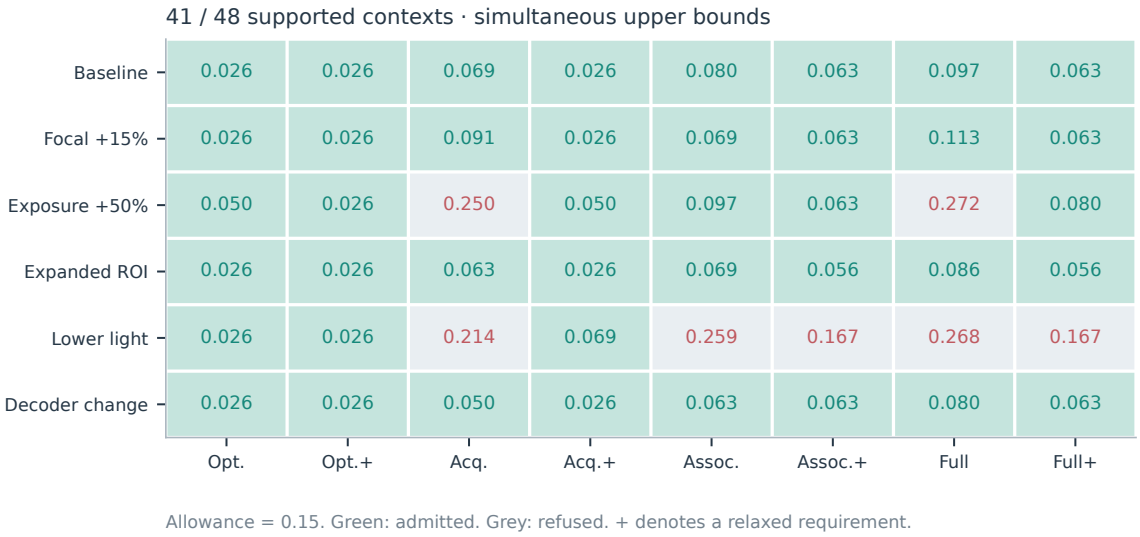


Figure 4: Calibrated event bounds for the controlled population. Green cells satisfy the primary allowance 0.15. The independent held-out check evaluates the same fixed programs after plan selection.

6.2. Renewal work and its attribution

Whole renewal constructs 48 cards; exact dependency caching constructs 42; containment with fallback and the independent rule implementation each construct 26. Their record evaluations are 12,288, 10,752 and 6,656, respectively. Containment reduces this work by 38.10% relative to exact caching while preserving all 41 supported contexts. Catalogue CPU is 4.419 ms for whole renewal, 4.290 ms for DAG exact, 2.725 ms for the compiled contract and 4.777 ms for the interpreted rules. The contract's absolute preparation saving over exact caching is 1.565 ms at this scale.

The 640 paired source-renewal cases consume 183.993 s of process CPU under whole renewal and 80.486 s under dependency caching, a 56.26% reduction. A paired bootstrap over the 16 timing inputs gives a 95% interval of 55.87–56.65%. Contract and rule paths consume 80.279 and 79.757 s. Their numerical outputs and stage counts agree with the DAG baseline. Figure 5 separates this shared numerical benefit from evidence preparation.

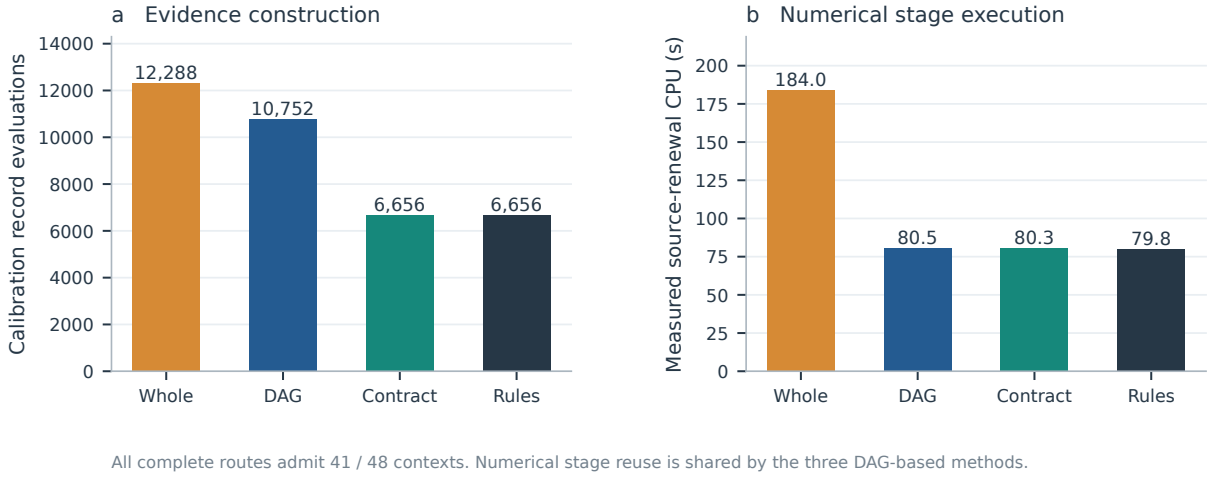


Figure 5: Two measured renewal costs. (a) Evidence construction on stored calibration records. (b) Actual numerical execution over 640 paired cases per method. Containment changes evidence work; the three dependency-based paths share numerical stage reuse.

6.3. Recoverable output attachment

Validated repair restores all 3,072 eligible fault scenarios, covering 1,536 distinct source/request contexts across the 256 images. The permanent latch restores none. The authoritative-snapshot and event-log implementations reproduce the validated-repair decisions exactly (Table 5). The source observation is retained throughout; the repair commits the trusted updated output at a newer revision.

The three recovery implementations issue no output with an incorrect numerical origin. The metadata-only ablation issues 3,072 such outputs, identifying the role of payload and provenance validation. Across all requests, 4,096 valid repairs commit and 24,576 malformed repair commands are rejected. A successfully repaired binding remains valid when a duplicate transaction is rejected. Strict complete service under exposure remains statistically inadmissible after repair; provenance recovery does not change its event bound.

Table 5: Output recovery on the same stored observations. Recovery denominator: 3,072 eligible scenarios. Wrong origin is checked against the archived source and producer record.

Method	Recovered	Wrong-origin issuances
Permanent latch	0	0
Authoritative snapshot	3,072	0
Validated repair	3,072	0
Event-log replay	3,072	0
Metadata-only ablation	3,072	3,072

6.4. Transfer to geometric surface rendering

The unchanged image front end processes all 72 path-traced images. Baseline panels have mean correct-address coverage 84.32% and median maximum mapping error 0.748 mm. At gain +1.3 stops, mean coverage

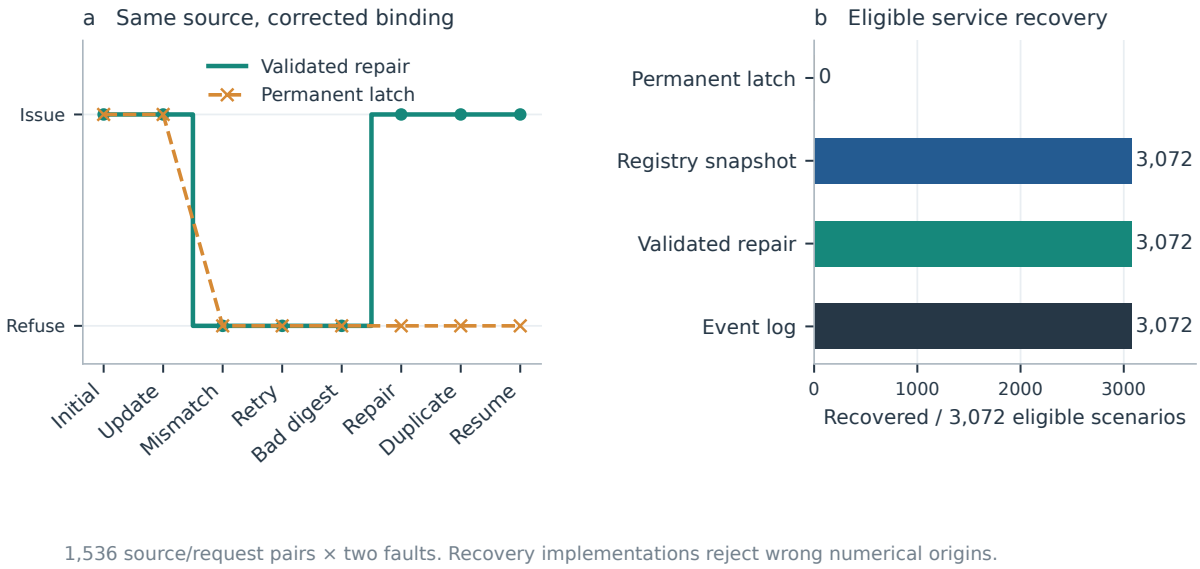


Figure 6: Recovery changes service availability while retaining the rejected attempt. (a) The first held-out source's relaxed complete-service workflow. (b) Recoverable scenarios under matched trusted-registry information. Numerical-origin checks and protocol traces use the same stored image outputs.

is 89.54%; at one-tenth illumination it is 78.12%. The inherited association predicate fails on 9/24 baseline, 5/24 gain and 10/24 lower-light panels (Fig. 7). Full scene-level outcomes retain every rendered input.

Blender's independently computed camera projections agree with the numerical projector within 8.5×10^{-5} pixel. The resulting comparison separates appearance-driven extraction behavior from a camera-coordinate mismatch. These transfer outcomes are scored within their own renderer population; the earlier 48-event calibration remains attached to its original producer and population.

6.5. Component regression and operating conditions

Conditional hidden couplings activate in 86 of 192 update/input pairs across the 64 regression inputs. Re-execution detects all 86 changes in quantities declared unaffected; all 192 clean pairs pass. The three fault classes activate on 24, 36 and 26 inputs. These checks test the declared dependency interface at component registration.

Stronger controlled disturbances lower mean coverage to 72.69% and raise the 95th-percentile mapping error to 13.095 mm. Strict complete-service failures rise from 4/256 nominal test scenes to 77/128 stress scenes (Fig. 8). Optical sampling remains acceptable, while occlusion, missing correspondences and CAD discrepancy affect the image-to-model task. This result identifies an operating-distribution change that requires its own evidence population.

7. Discussion

The engineering contribution is a decision procedure for keeping inspection evidence applicable through change. Lifecycle knowledge graphs establish how versions and operating states remain connected [6]; dynamic safety cases establish how an evolving system can trigger evidence updates [7]. Here the maintained object is an explicit statistical statement about a requested quality event. Its use requires a compatible population and producer, a covering event bound, and a current output binding. These conditions turn a change history into a concrete retain, renew, refuse or recover decision.

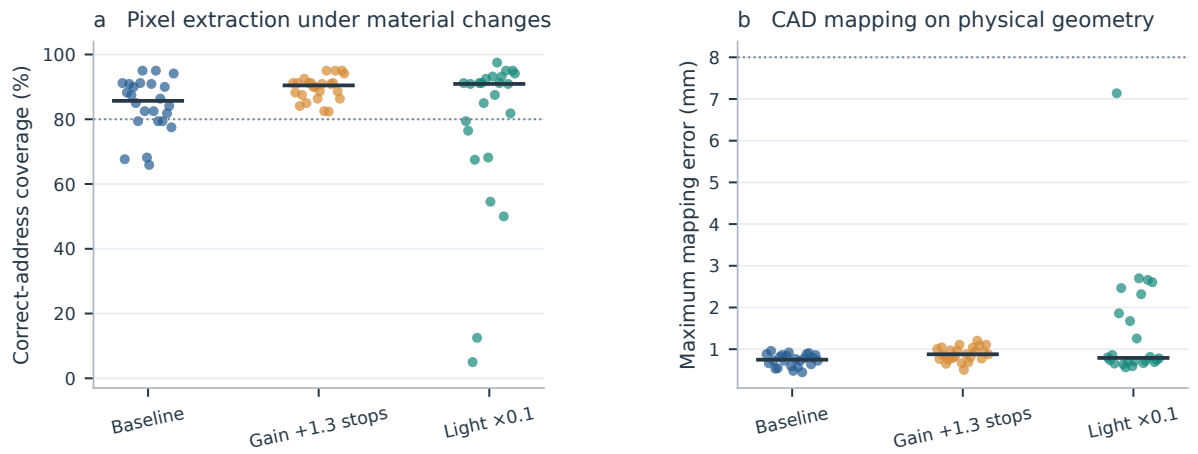


Figure 7: Image-to-CAD transfer across 24 independently rendered panels. Every scene is processed in all three configurations. The unchanged detector receives image pixels; physical CAD locations enter subsequent scoring. Dashed lines show inherited association thresholds.

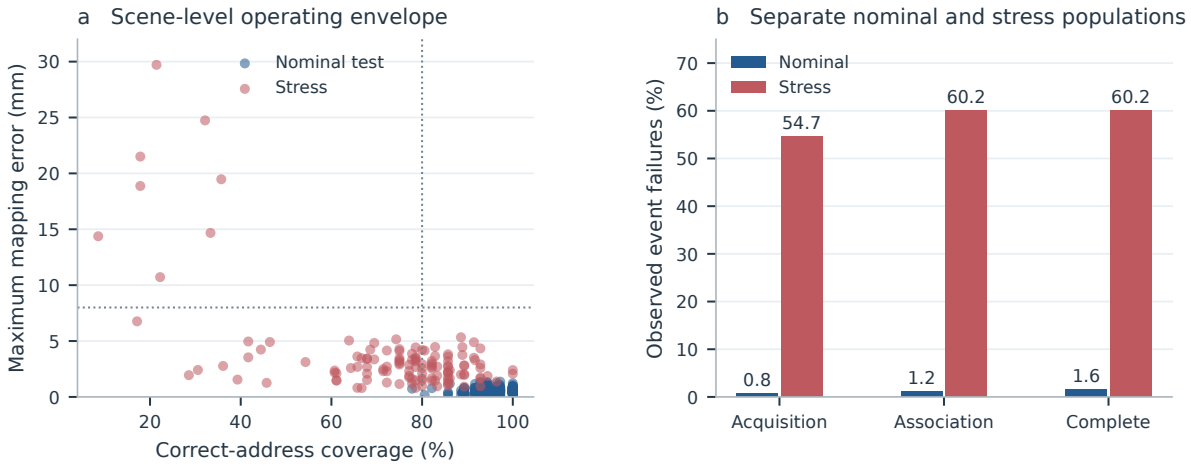


Figure 8: Controlled nominal and stress populations. Each point represents one independently sampled complete scene. Stress changes missingness, occlusion, CAD discrepancy and prior error; the resulting failures are reported separately from nominal calibration.

The matched comparisons identify the source of each benefit. Dependency caching supplies the large numerical saving, consistent with incremental computation and selective recomputation [17, 15]. Containment reduces evidence-record evaluations by 38.10% while exact fallback preserves all 41 supported contexts. At the present scale, the corresponding preparation saving is 1.565 ms. The further engineering value is explicit applicability: a relaxed request can reuse a stronger statement, a changed quality function triggers renewal, and a corrected output can recover its binding without changing the event's risk bound. Authoritative-snapshot and event-log implementations reproduce these decisions, showing which behavior follows from the interface semantics.

Provenance and numerical quality answer different inspection questions. Generation and revision records trace the source of a result [8]; a trusted registry authenticates the current attachment. Fitted residual describes agreement with extracted centers, whereas physical mapping error evaluates the assigned CAD address. The running example demonstrates their interaction: correcting a stale output restores provenance, but strict complete service after the exposure change remains inadmissible. Service under the relaxed request resumes only when its own calibrated bound permits it.

The two image populations clarify the role of the numerical provider. Controlled scenes support paired changes and an independent calibration/test experiment. The path-traced panels challenge the unchanged detector with geometric fasteners, reflections and altered illumination. Baseline association fails on 9 of 24 such panels, identifying a concrete front-end transfer issue. In ultrasonic concrete monitoring, simulation-to-experiment adaptation is evaluated with both simulated and experimental signals [26]. Accordingly, each renderer has its own population identity, and its outcomes remain attached to that population.

The implemented workflow operates with a trusted registry and declared producer dependencies. Paired regression can expose a broken dependency declaration on executed inputs; provider registration also requires maintaining the declared interface. These assumptions define the operating scope, and the archived records make each decision inspectable. For engineering use, the central result is the alignment of the requested quality, its calibrated producer and the output to which the evidence is attached.

8. Conclusion

Executable evidence management connects a quality requirement to the calibrated producer and authenticated output that support it. Containment with exact fallback preserves direct-renewal support, while validated repair restores a corrected binding and retains its audit history. The controlled image study supports 41 service contexts, independently checks their held-out risks and reduces evidence-record evaluations by 38.10%. Matched implementations separate this evidence benefit from conventional numerical caching and reproduce recovery across 1,536 eligible source/request contexts. The geometric-rendering and stress studies expose the role of appearance and operating-population changes. The resulting interface provides a concrete specification for selective renewal and recoverable evidence attachment in CAD-guided inspection.

References

[1] I. Jovančević, I. Viana, J.-J. Orteu, T. Sentenac, S. Larnier, Matching CAD model and image features for robot navigation and inspection of an aircraft, in: Proceedings of the 5th International Conference on Pattern Recognition Applications and Methods, 2016, pp. 359–366. doi:10.5220/0005756303590366.

[2] O. S. Amosov, S. G. Amosova, I. O. Iochkov, Deep neural network recognition of rivet joint defects in aircraft products, Sensors 22 (9) (2022) 3417. doi:10.3390/s22093417.

[3] S. Ma, K. A. Flanigan, M. Bergés, State-of-the-art review and synthesis: A requirement-based roadmap for standardized predictive maintenance automation using digital twin technologies, Advanced Engineering Informatics 62 (2024) 102800. doi:10.1016/j.aei.2024.102800.

[4] K. Hu, C. Yan, C. Zhou, T. Deng, L. Wang, Y. Xu, C. Zhuang, Adaptive threshold optimization for efficient and robust high-dimensional sensor calibration in building energy systems, Advanced Engineering Informatics 77 (2027) 105365, available online 10 October 2026. doi:10.1016/j.aei.2026.105365.

[5] Q. Zou, H.-Y. Feng, A decision-support method for information inconsistency resolution in direct modeling of CAD models, Advanced Engineering Informatics 44 (2020) 101087. doi:10.1016/j.aei.2020.101087.

[6] C. Yang, Y. Liu, H. Dong, P. Zhang, J. Lu, B. Zhu, R. Ojala, K. Tammi, Lifecycle-aware knowledge graphs with human-in-the-loop agentic retrieval for automotive manufacturing, Advanced Engineering Informatics 76 (2026) 105122. doi:10.1016/j.aei.2026.105122.

[7] E. Denney, G. Pai, I. Habli, Dynamic safety cases for through-life safety assurance, in: 2015 IEEE/ACM 37th IEEE International Conference on Software Engineering, IEEE, 2015, pp. 587–590. doi:10.1109/ICSE.2015.199.

[8] L. Moreau, P. Missier, PROV-DM: The PROV data model, W3C recommendation, World Wide Web Consortium, editors; Recommendation of 30 April 2013 (Apr. 2013).
URL https://www.w3.org/TR/2013/REC-prov-dm-20130430/

[9] L. Petnga, M. Austin, An ontological framework for knowledge modeling and decision support in cyber-physical systems, Advanced Engineering Informatics 30 (1) (2016) 77–94. doi:10.1016/j.aei.2015.12.003.

[10] C. Quinn, J. J. McArthur, A case study comparing the completeness and expressiveness of two industry recognized ontologies, Advanced Engineering Informatics 47 (2021) 101233. doi:10.1016/j.aei.2020.101233.

[11] L. Ma, T. Hartmann, A proposed ontology to support the hardware design of building inspection robot systems, Advanced Engineering Informatics 55 (2023) 101851. doi:10.1016/j.aei.2022.101851.

[12] J. Xiao, Q. Wang, Y. Li, Z. Yu, H. Zhou, B. Lin, A fully automated DM-BIM-BEM pipeline enabling graph-based intelligence, interoperability, and performance-driven early design, Advanced Engineering Informatics 75 (2026) 104824. doi:10.1016/j.aei.2026.104824.

[13] J. Cheney, P. Missier, L. Moreau, Constraints of the PROV data model, W3C recommendation, World Wide Web Consortium, editors; Recommendation of 30 April 2013; contributing author Tom De Nies (Apr. 2013).
URL https://www.w3.org/TR/2013/REC-prov-constraints-20130430/

[14] P. Missier, S. Woodman, H. Hiden, P. Watson, Provenance and data differencing for workflow reproducibility analysis, Concurrency and Computation: Practice and Experience 28 (4) (2016) 995–1015. doi:10.1002/cpe.3035.

[15] P. Missier, J. Cała, M. Rathi, Preserving the value of large scale data analytics over time through selective re-computation, in: Data Analytics, Vol. 10365 of Lecture Notes in Computer Science, Springer, 2017, pp. 65–77. doi:10.1007/978-3-319-60795-5_6.

[16] J. Cała, P. Missier, Provenance annotation and analysis to support process re-computation, in: Provenance and Annotation of Data and Processes, Vol. 11017 of Lecture Notes in Computer Science, Springer, 2018, pp. 3–15. doi:10.1007/978-3-319-98379-0_1.

[17] A. Mokhov, N. Mitchell, S. Peyton Jones, Build systems à la carte, Proceedings of the ACM on Programming Languages 2 (ICFP) (2018) 1–29. doi:10.1145/3236774.

[18] S. Martínez-Sanchis, SafeMachine: An evidence-centric architecture for traceable and context-aware machine safety assessment, Advanced Engineering Informatics 77 (2027) 105321, available online in 2026. doi:10.1016/j.aei.2026.105321.

[19] A. Benveniste, B. Caillaud, D. Nickovic, R. Passerone, J.-B. Raclet, P. Reinkemeier, A. Sangiovanni-Vincentelli, W. Damm, T. A. Henzinger, K. G. Larsen, Contracts for system design, Foundations and Trends in Electronic Design Automation 12 (2–3) (2018) 124–400. doi:10.1561/1000000053.

[20] U. Mehmood, S. Sheikhi, S. Bak, S. A. Smolka, S. D. Stoller, The black-box simplex architecture for runtime assurance of autonomous CPS, in: NASA Formal Methods, Vol. 13260 of Lecture Notes in Computer Science, Springer, 2022, pp. 231–250. doi:10.1007/978-3-031-06773-0_12.

[21] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, The Annals of Applied Statistics 19 (2) (2025) 1641–1662. doi:10.1214/24-AOAS1998.

[22] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster, Conformal risk control, in: The Twelfth International Conference on Learning Representations, 2024.
URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/f3549ef9b5ff520a7e41ff3cc306ab2b-Abstract-Conference.html

[23] C. J. Clopper, E. S. Pearson, The use of confidence or fiducial limits illustrated in the case of the binomial, Biometrika 26 (4) (1934) 404–413. doi:10.1093/biomet/26.4.404.

[24] T. Liu, B. Li, X. Du, B. Jiang, X. Jin, L. Jin, Z. Zhao, Component-aware anomaly detection framework for adjustable and logical industrial visual inspection, Advanced Engineering Informatics 58 (2023) 102161. doi:10.1016/j.aei.2023.102161.

[25] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, P. Abbeel, Domain randomization for transferring deep neural networks from simulation to the real world, in: 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017, pp. 23–30. doi:10.1109/iros.2017.8202133.

[26] C. Xu, G. Vu, B. T. Cao, Z. Liu, F. Diewald, Y. Yuan, G. Meschke, Bridging simulation and experiment: A self-supervised domain adaptation framework for concrete damage classification, Advanced Engineering Informatics 69 (2026) 104125. doi:10.1016/j.aei.2025.104125.

[27] R. Hartley, A. Zisserman, Multiple View Geometry in Computer Vision, 2nd Edition, Cambridge University Press, 2004. doi:10.1017/CBO9780511811685.

[28] D. F. Crouse, On implementing 2D rectangular assignment algorithms, IEEE Transactions on Aerospace and Electronic Systems 52 (4) (2016) 1679–1696. doi:10.1109/TAES.2016.140952.

[29] P. J. Huber, Robust estimation of a location parameter, The Annals of Mathematical Statistics 35 (1) (1964) 73–101. doi:10.1214/aoms/1177703732.

[30] J. T. Kajiya, The rendering equation, ACM SIGGRAPH Computer Graphics 20 (4) (1986) 143–150. doi:10.1145/15886.15902.

[31] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, SciPy 1.0 Contributors, SciPy 1.0: fundamental algorithms for scientific computing in Python, Nature Methods 17 (3) (2020) 261–272. doi:10.1038/s41592-019-0686-2.